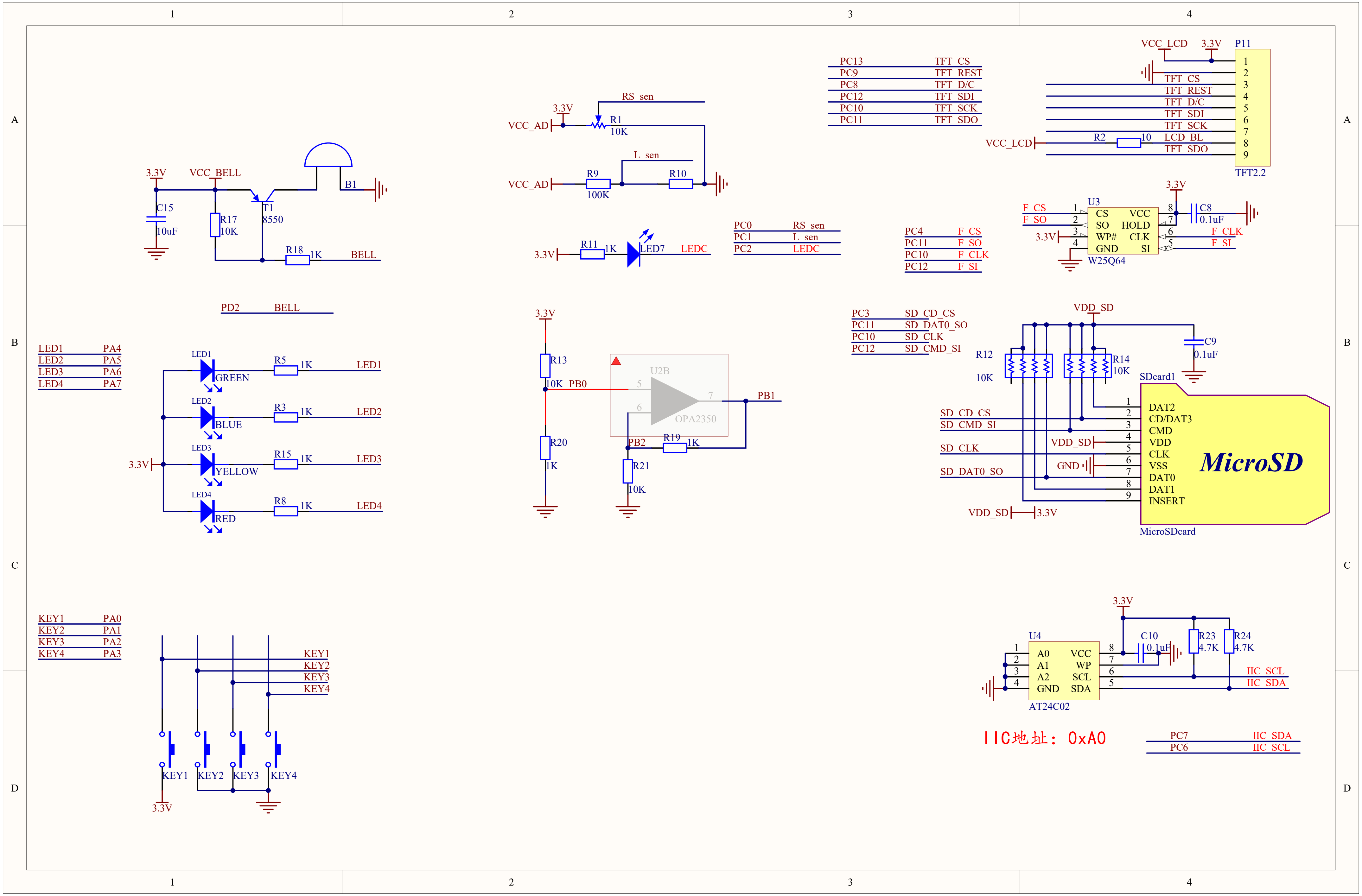
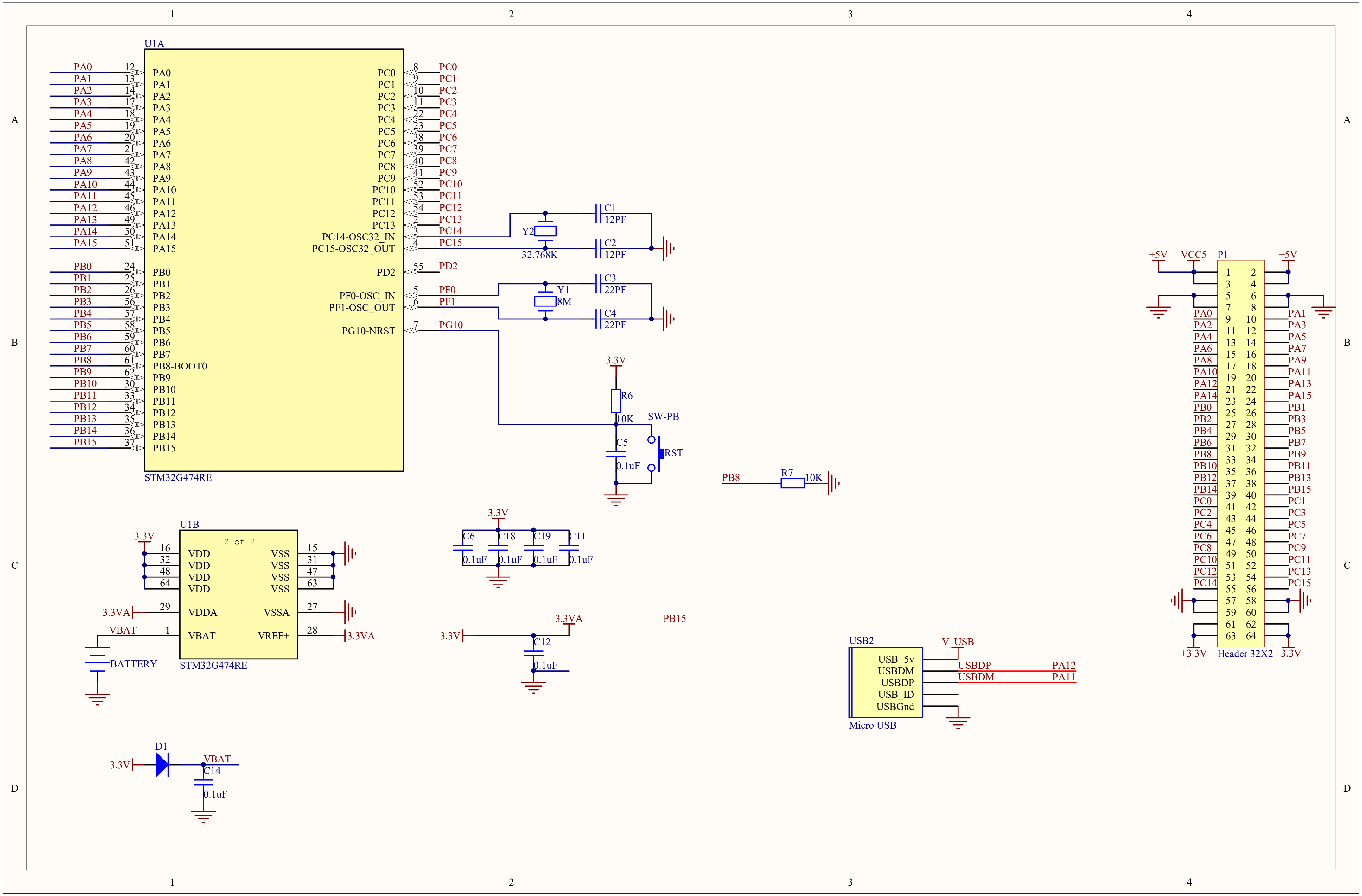
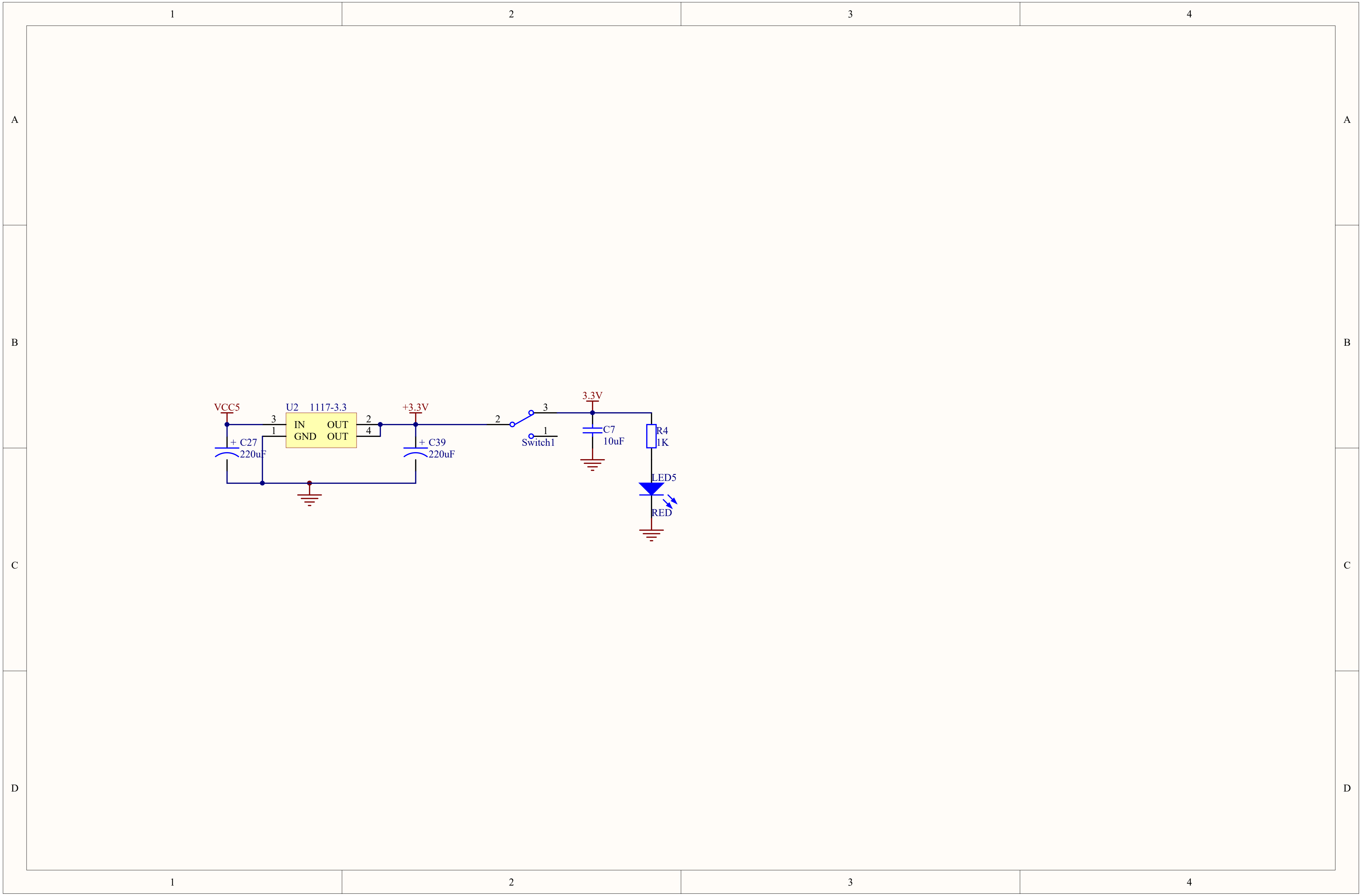
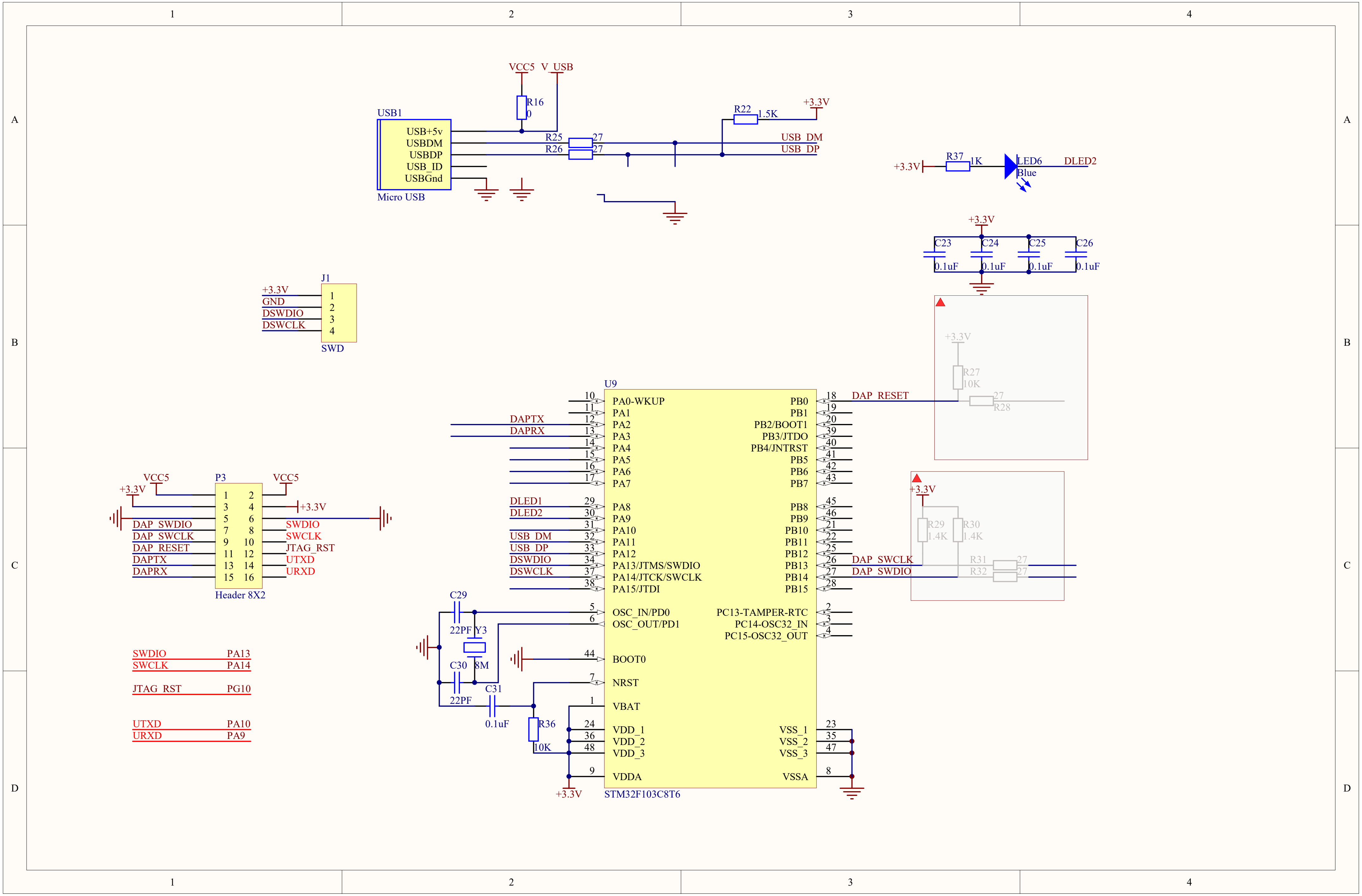
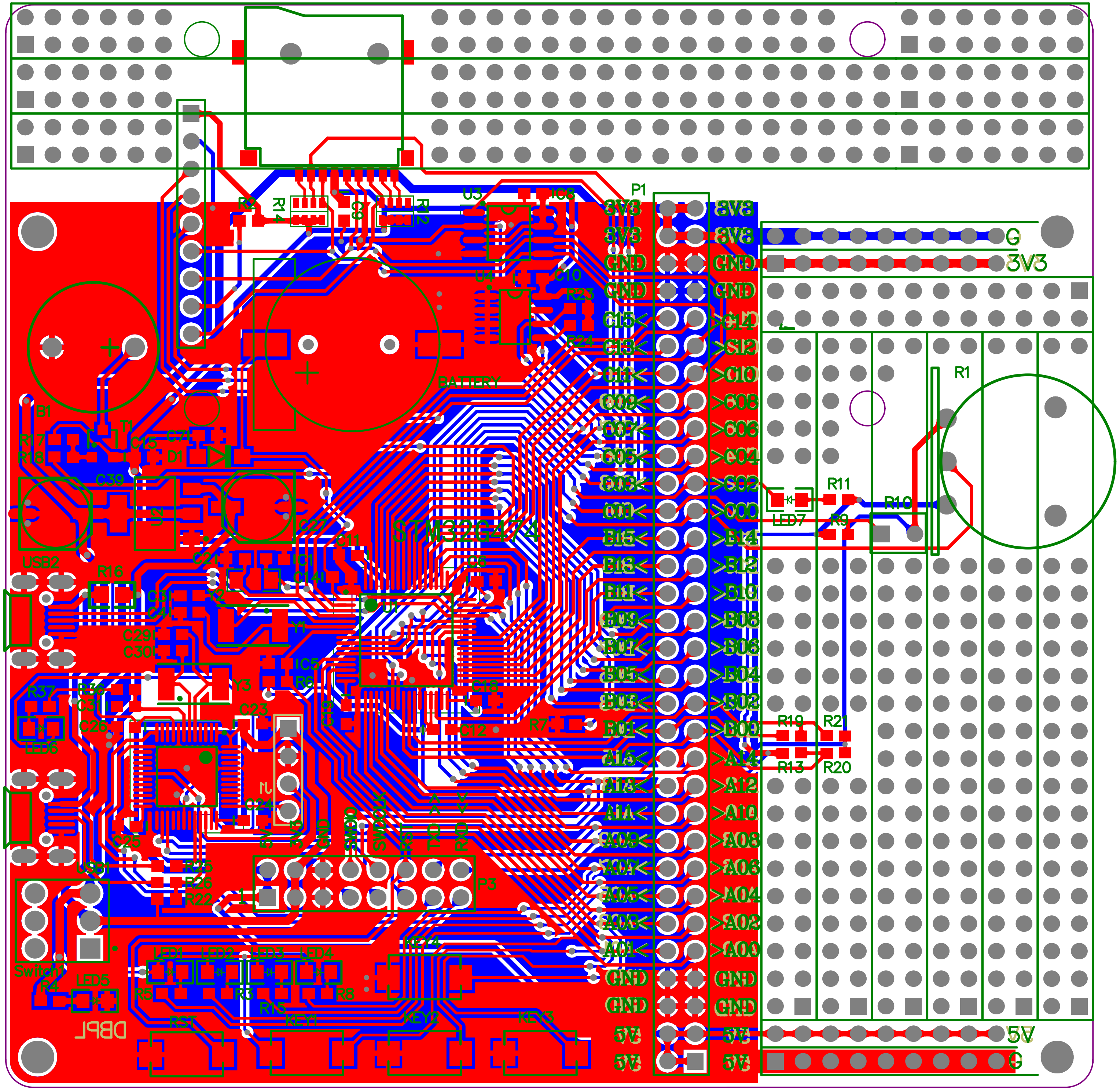

1
2
3
4
A
B
C
D
3.3V
VCC_BELL
C15
10uF
R17
10K
T1
8550
B1
R18
1K
BELL
PD2
BELL
VCC_AD
3.3V
RS_sen
R1
10K
L_sen
R9
100K
R10
3.3V
R11
1K
LED7
LEDC
PC13
TFT_CS
PC9
TFT_REST
PC8
TFT_D/C
PC12
TFT_SDI
PC10
TFT_SCK
PC11
TFT_SDO
PC0
RS_sen
PC1
L_sen
PC2
LEDC
PC4
F_CS
PC11
F_SO
PC10
F_CLK
PC12
F_SI
VCC_LCD
3.3V
P11
TFT_CS
TFT_REST
TFT_D/C
TFT_SDI
TFT_SCK
LCD_BL
TFT_SDO
VCC_LCD
R2
10
1
2
3
4
5
6
7
8
9
TFT2.2
3.3V
U3
F_CS
F_SO
3.3V
CS
SO
WP#
GND
VCC
HOLD
CLK
SI
C8
0.1uF
F_CLK
F_SI
W25Q64
LED1
PA4
LED2
PA5
LED3
PA6
LED4
PA7
LED1
GREEN
R5
1K
LED1
LED2
BLUE
R3
1K
LED2
3.3V
LED3
YELLOW
R15
1K
LED3
LED4
RED
R8
1K
LED4
3.3V
R13
10K
PB0
R20
1K
U2B
5
6
7
OPA2350
PB1
PB2
R19
1K
R21
10K
PC3
SD_CD_CS
PC11
SD_DAT0_SO
PC10
SD_CLK
PC12
SD_CMD_SI
VDD_SD
C9
0.1uF
R12
10K
R14
10K
SDcard1
SD_CD_CS
SD_CMD_SI
VDD_SD
SD_CLK
GND
SD_DAT0_SO
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
INSERT
MicroSD
VDD_SD
3.3V
MicroSDcard
KEY1
PA0
KEY2
PA1
KEY3
PA2
KEY4
PA3
KEY1
KEY2
KEY3
KEY4
KEY1
KEY2
KEY3
KEY4
3.3V
3.3V
U4
A0
A1
A2
GND
VCC
WP
SCL
SDA
C10
0.1uF
R23
4.7K
R24
4.7K
IIC_SCL
IIC_SDA
AT24C02
PC7
IIC_SDA
PC6
IIC_SCL

IIC地址：0xA0


1
2
3
4
A
B
C
D
U1A
PA0 12 PA0
PA1 13 PA1
PA2 14 PA2
PA3 17 PA3
PA4 18 PA4
PA5 19 PA5
PA6 20 PA6
PA7 21 PA7
PA8 42 PA8
PA9 43 PA9
PA10 44 PA10
PA11 45 PA11
PA12 46 PA12
PA13 49 PA13
PA14 50 PA14
PA15 51 PA15
PB0 24 PB0
PB1 25 PB1
PB2 26 PB2
PB3 56 PB3
PB4 57 PB4
PB5 58 PB5
PB6 59 PB6
PB7 60 PB7
PB8 61 PB8-BOOT0
PB9 62 PB9
PB10 30 PB10
PB11 33 PB11
PB12 34 PB12
PB13 35 PB13
PB14 36 PB14
PB15 37 PB15
PC0 8 PC0
PC1 9 PC1
PC2 10 PC2
PC3 11 PC3
PC4 22 PC4
PC5 23 PC5
PC6 38 PC6
PC7 39 PC7
PC8 40 PC8
PC9 41 PC9
PC10 52 PC10
PC11 53 PC11
PC12 54 PC12
PC13 2 PC13
PC14-OSC32_IN 3 PC14
PC15-OSC32_OUT 4 PC15
PD2 55 PD2
PF0-OSC_IN 5 PF0
PF1-OSC_OUT 6 PF1
PG10-NRST 7 PG10
STM32G474RE
C1
12PF
Y2
32.768K
C2
12PF
C3
22PF
Y1
8M
C4
22PF
3.3V
R6
10K
SW-PB
C5
0.1uF
RST
PB8
R7
10K
U1B
2 of 2
3.3V
16 VDD
32 VDD
48 VDD
64 VDD
VSS 15
VSS 31
VSS 47
VSS 63
3.3VA
29 VDDA
VSSA 27
VBAT
1 VBAT
VREF+ 28
3.3VA
BATTERY
STM32G474RE
3.3V
C6
0.1uF
C18
0.1uF
C19
0.1uF
C11
0.1uF
3.3VA
3.3V
C12
0.1uF
PB15
D1
3.3V
VBAT
C14
0.1uF
USB2
USB+5v
USBDM
USBDP
USB_ID
USBGnd
Micro USB
V_USB
USBDP
PA12
USBDM
PA11
+5V
VCC5
P1
+5V
1 2
3 4
5 6
7 8
PA0 9 10 PA1
PA2 11 12 PA3
PA4 13 14 PA5
PA6 15 16 PA7
PA8 17 18 PA9
PA10 19 20 PA11
PA12 21 22 PA13
PA14 23 24 PA15
PB0 25 26 PB1
PB2 27 28 PB3
PB4 29 30 PB5
PB6 31 32 PB7
PB8 33 34 PB9
PB10 35 36 PB11
PB12 37 38 PB13
PB14 39 40 PB15
PC0 41 42 PC1
PC2 43 44 PC3
PC4 45 46 PC5
PC6 47 48 PC7
PC8 49 50 PC9
PC10 51 52 PC11
PC12 53 54 PC13
PC14 55 56 PC15
57 58
59 60
61 62
63 64
+3.3V
Header 32X2
+3.3V


1
2
3
4
A
B
C
D
VCC5
U2
1117-3.3
+3.3V
3
IN
OUT
2
1
GND
OUT
4
+ C27
220uF
+ C39
220uF
2
3
1
Switch1
3.3V
C7
10uF
R4
1K
LED5
RED


VCC5
V_USB
R16
0
USB1
USB+5v
USBDM
USBDP
USB_ID
USBGnd
Micro USB
R25
27
R26
27
R22
1.5K
+3.3V
USB_DM
USB_DP
+3.3V
R37
1K
LED6
Blue
DLED2
+3.3V
C23
0.1uF
C24
0.1uF
C25
0.1uF
C26
0.1uF
J1
+3.3V
GND
DSWDIO
DSWCLK
1
2
3
4
SWD
+3.3V
R27
10K
R28
27
DAP_RESET
U9
PA0-WKUP
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13/JTMS/SWDIO
PA14/JTCK/SWCLK
PA15/JTDI
OSC_IN/PD0
OSC_OUT/PD1
BOOT0
NRST
VBAT
VDD_1
VDD_2
VDD_3
VDDA
PB0
PB1
PB2/BOOT1
PB3/JTDO
PB4/JNTRST
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC13-TAMPER-RTC
PC14-OSC32_IN
PC15-OSC32_OUT
VSS_1
VSS_2
VSS_3
VSSA
STM32F103C8T6
DAPTX
DAPRX
DLED1
DLED2
USB_DM
USB_DP
DSWDIO
DSWCLK
+3.3V
R29
1.4K
R30
1.4K
R31
27
R32
27
DAP_SWCLK
DAP_SWDIO
VCC5
+3.3V
P3
VCC5
+3.3V
DAP_SWDIO
DAP_SWCLK
DAP_RESET
DAPTX
DAPRX
SWDIO
SWCLK
JTAG_RST
UTXD
URXD
Header 8X2
C29
22PF
Y3
8M
C30
22PF
C31
0.1uF
R36
10K
+3.3V
SWDIO PA13
SWCLK PA14
JTAG_RST PG10
UTXD PA10
URXD PA9
1
2
3
4
A
B
C
D


P1
3V3
3V3
GND
GND
C15<
C13<
C11<
C09<
C07<
C05<
C03<
C01<
B15<
B13<
B11<
B09<
B07<
B05<
B03<
B01<
A15<
A13<
A11<
A09<
A07<
A05<
A03<
A01<
GND
GND
5V
5V
3V3
3V3
GND
GND
>C14
>C12
>C10
>C08
>C06
>C04
>C02
>C00
>B14
>B12
>B10
>B08
>B06
>B04
>B02
>B00
>A14
>A12
>A10
>A08
>A06
>A04
>A02
>A00
GND
GND
5V
5V
G
3V3
5V
G
U3
C6
R13
R14
C9
R12
U4
C10
R23
R24
BATTERY
B1
R17
R18
C39
D1
USB2
R16
X1
Y1
Y3
IC5
R6
R5
C23
C28
R37
LED6
J1
SWD
SWCLK
SWDIO
P3
R1
R11
R10
R9
LED7
R19
R21
R13
R20
LED1
LED2
LED3
LED4
LED5
R2
R3
R15
R8
KEY1
KEY2
KEY3
RST
SW1
DBPL